

A probability experiment

A chance experiment produces an uncertain result called an outcome.

What are the possible outcomes when a fair six-sided die is rolled?

Sample space

The sample space S contains all possible outcomes of the experiment.

What sample space describes two ordered tosses of a fair coin?

An event

An event is a set of outcomes from the sample space.

For a die roll, what outcomes belong to “an even number”?

Equally likely outcomes

Equally likely outcomes have the same probability. A fair die has six equally likely faces.

What is the probability of a value greater than 4 on a fair six-sided die?

Unequal probabilities

An unfair or biased device gives unequal chances to outcomes intended to be equally likely. A spinner has regions of sizes $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{4}$.

Are its three color outcomes equally likely?

Probability bounds

An event probability lies between 0 and 1.

What do probabilities 0 and 1 mean?

Relative frequency

Empirical means observed. Relative frequency is an observed count divided by trials.

Synthetic simulation: 124 heads in 250 tosses.

What is the observed relative frequency of heads?

Law of large numbers

As independent repetitions increase, long-term relative frequency tends toward theoretical probability. A fair coin has $P(\text{heads}) = 0.5$.

Must the first ten tosses include exactly five heads?

A complement

The complement of event A contains outcomes outside A. OpenStax writes A' ; A^c means the same event.

$$P(A^c) = 1 - P(A)$$

If $P(A) = 0.35$, what is $P(A\text{-complement})$?

Translate “not”

A fair six-sided die is rolled. A is “roll a 6.”

What outcomes belong to A-complement?

The intersection

The AND event, or intersection $A \cap B$, contains outcomes in both A and B.

For a fair die, $A = \text{even}$ and $B = \text{greater than 3}$. Which outcomes belong to $A \text{ AND } B$?

The union

The OR event, or union $A \cup B$, contains outcomes in A, B, or both.

For a fair die, A = even and B = greater than 3. Which outcomes belong to A OR B ?

Inclusive “or”

One card is drawn from a standard 52-card deck.

Does “a heart or a king” include the king of hearts?

Conditional probability

$P(A | B)$ is the probability of A when the outcome is restricted to B.

For a fair die, what is $P(\text{even} | \text{greater than } 3)$?

The conditional-probability formula

Condition on B only when $P(B) > 0$.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

**For $P(A \text{ AND } B) = 0.12$ and $P(B) = 0.4$,
what is $P(A \mid B)$?**

The conditioned group

Synthetic survey: 20 students use tablets; 8 of those also use laptops. Select one student uniformly from the tablet users.

What is the denominator for $P(\text{laptop} | \text{tablet})$?

Reverse the condition

Synthetic survey: 20 tablet users, 30 laptop users, and 8 use both. Select one of these surveyed students uniformly.

Is $P(\text{laptop} \mid \text{tablet})$ equal to $P(\text{tablet} \mid \text{laptop})$?

Without replacement

A box holds 3 red and 2 blue tokens. Draw one token, keep it, then draw another.

How does the first draw affect the possible second draw?

Write a probability event

Let X be the number of heads in three independent fair-coin tosses.

How would you write “at least two heads” using X ?